Labor Demand

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UBC

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- Theory of the firm $\rightarrow$ Theory of labor demand $\rightarrow$ Theory of Inequality
  - Demand for inputs (e.g. labor, capital) from optimization.
  - Key question: what explains the elasticity of demand for labor?
- Today: neo-classical labor demand.
- Building block for a variety of "applied" questions:
  - Effects of technology
  - Effects of immigration / demographics.
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  - Some of these will populate the "Inequality" wars chapter.

The building block

- Firms produce an output combining multiple inputs:

$$y = f(x_1, x_2, \ldots, x_N) \tag{1}$$

- Typical assumptions:
  - $f(.)$ exhibits constant return to scale (CRS).
  - work with only two inputs: $y = f(x_1, x_2)$.

- Notation time-out
  - $f_i \equiv$ partial derivative of f with respect to x_i with $i \in \{1, 2\}$.

An Old friend

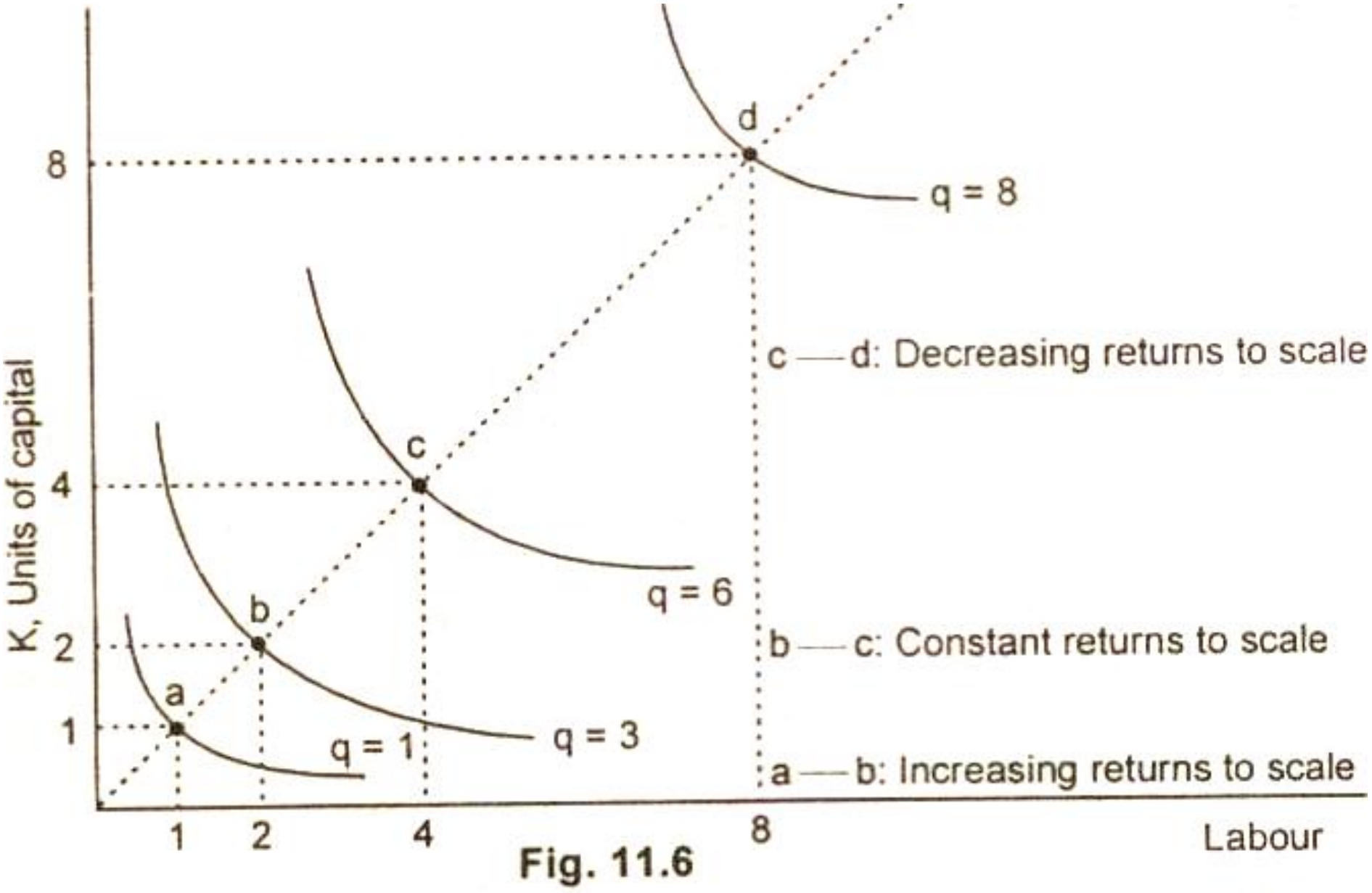


Fig. 11.6

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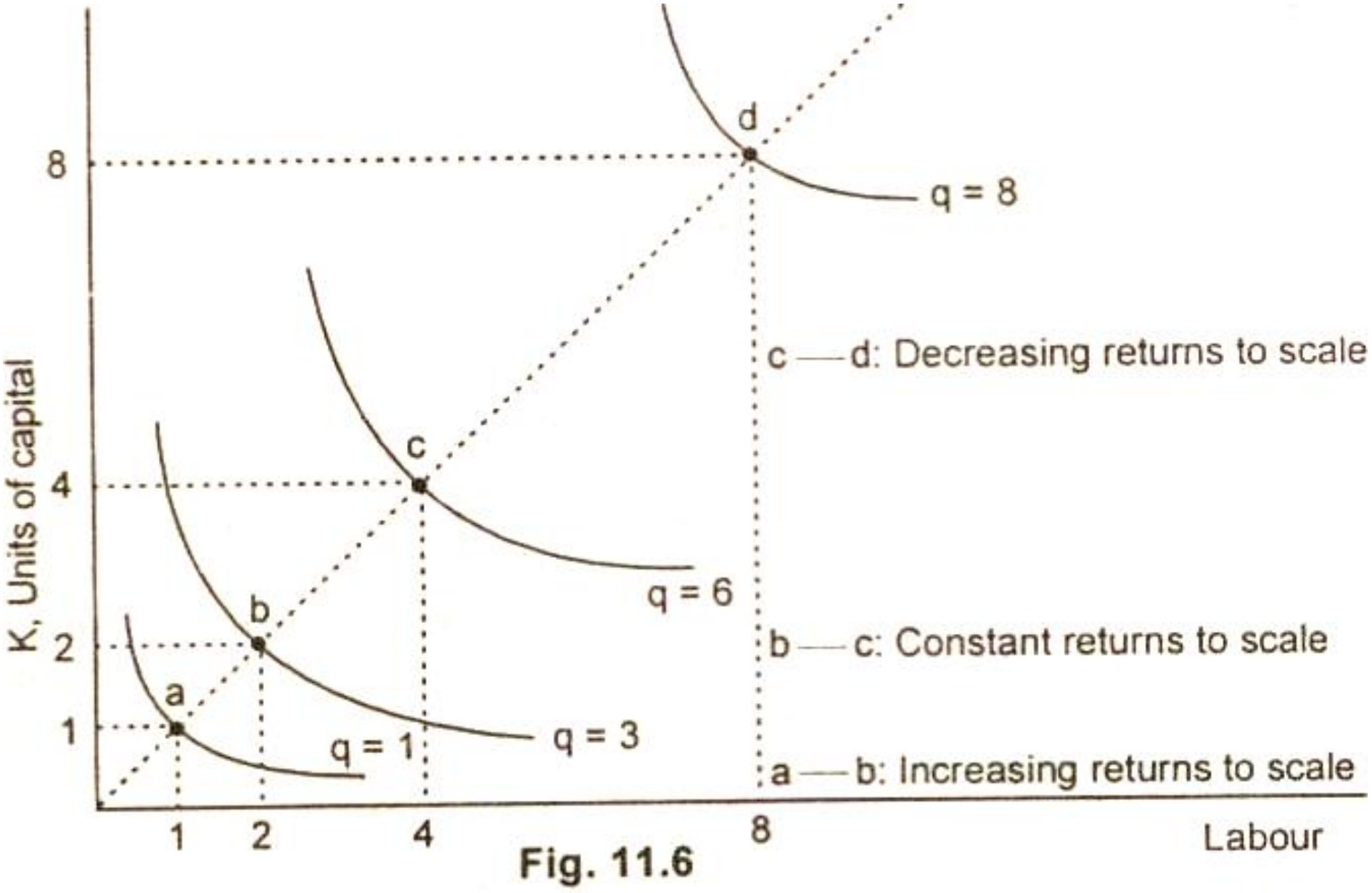


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The Holy Grail: Elasticity of substitution

- Under CRS and two inputs: shape of the isoquant is going to be given by

$$\sigma = -\frac{dlog(\frac{x_1}{x_2})}{dlog(\frac{f_1}{f_2})} \tag{2}$$

- Pop-quiz

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- Name of the game in labor demand (most of the time): estimate σ

Special Cases

- Cobb-Douglas

$$y = f(x_1, x_2) = x_1^{\alpha} x_2^{1-\alpha} \tag{3}$$

$\sigma = 1$

- CES:

$$y = f(x_1, x_2) = (\alpha x_1^{-\rho} + (1-\alpha) x_2^{-\rho})^{-\frac{1}{\rho}} \tag{4}$$

$$\sigma = \frac{1}{(1+\rho)}$$

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- Sheppard's Lemma (Envelope Theorem):

$$x_1 = C_1(w_1, w_2, y). \tag{6}$$

$$x_2 = C_2(w_1, w_2, y). \tag{7}$$

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- Important: all derivations that are coming up are for a fixed output y.

General case

- We have that C_i is homogenous of degree 0 (HD_0) , then

$$x_1 = C_1(\frac{w_1}{w_2}, 1, y) \equiv g(\frac{w_1}{w_2}, y) \tag{8}$$

$$x_2 = C_2(\frac{w_1}{w_2}, y) \equiv h(\frac{w_1}{w_2}, y) \tag{9}$$

- Then we have that

$$C_{11} = \frac{\partial x_1}{\partial w_1} = \frac{1}{w_2} g_1(\frac{w_1}{w_2}, y) \tag{10}$$

$$C_{21} = \frac{\partial x_2}{\partial w_1} = \frac{1}{w_2} h_1(\frac{w_1}{w_2}, y) \tag{11}$$

General Case

- Then, for a fixed y,

$$log\frac{x_1}{x_2} = \log g(\frac{w_1}{w_2}, y) - \log h(\frac{w_1}{w_2}, y) \tag{12}$$

- Putting the pieces together

$$\begin{aligned}\sigma &= -\frac{\left(\frac{w_1}{w_2}\right) d\log(\frac{x_1}{x_2})}{d\frac{w_1}{w_2}} = -\frac{w_1}{w_2}\left(\frac{g_1(\frac{w_1}{w_2}, y)}{g} - \frac{h_1(\frac{w_1}{w_2}, y)}{h}\right) \\ &= -\frac{w_1}{w_2}\left(\frac{w_2 C_{11}}{g} - \frac{C_{21} w_2}{h}\right) \\ &\equiv -\frac{w_1 C_{11}}{C_1} + \frac{w_1 C_{21}}{C_2}\end{aligned} \tag{13}$$

General Case

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- Now apply Euler's theorem to the function C_j (HD_0)

$$C_{11}w_1 + C_{12}w_2 = 0 \tag{14}$$

- Based on the above, we arrive at

$$\sigma = \frac{w_1 C_{21}}{C_2} + \frac{w_2 C_{12}}{C_1} = C_{12}\frac{(w_1 C_1 + w_2 C_2)}{C_1 C_2} \tag{15}$$

- Applying Euler's theorem to $C(w_1, w_2)$ (which recall is a function HD_1), that is

$$w_1 C_1 + w_2 C_2 = C(w_1, w_2) \tag{16}$$

- We finally obtain

$$\sigma = \frac{C_{12}C}{C_1 C_2} \tag{17}$$

- Extension: partial elasticity of substitution multi-factor production functions

$$\sigma_{ij} = \frac{C_{ij}C}{C_i C_j} \tag{18}$$

Connecting the dots..

- Why these derivations are useful? Because we can start to see the importance of σ in driving quantities that we care about...

$$
\begin{aligned}
\frac{\partial x_1}{\partial w_2} &= C_{12} \\
\rightarrow \epsilon_{12} &\equiv \frac{w_2}{x_1}\frac{\partial x_1}{\partial w_2} = \frac{w_2 C_{12}}{x_1} \\
&= \frac{C_{12} C}{C_1 C_2}\frac{w_2 C_2 C_1}{C x_1} \\
&= \sigma \frac{w_2 x_2}{C} \\
&= \sigma \underbrace{s_2}_{\text{Input 2 cost share}}
\end{aligned}
\tag{19}
$$

- Also, we have that

$$
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w_1 \frac{\partial x_1}{\partial w_1} + w_2 \frac{\partial x_1}{\partial w_2} &= 0 \\
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- Important reminder: we were holding y fixed thus far!

Marshall's Laws

- We are now ready to derive Marshall's Laws for labor demand.
- These laws are derived under the following setup:
  - Perfectly competitive industry with CRS and cost function $C(w, y) = \mu(w)y$ where $w = (w_1, \ldots, w_N)$.
  - Output is priced at $p = \mu(w)$.
  - Downward sloping demand curve for the industry's output $y = D(p)$.
- Next, we will show how the elasticity of demand for an input x_i relates to three key parameters
  1. $\eta \equiv$ elasticity of product demand.
  2. $\sigma_{ij} \equiv$ partial elasticity of substitution.
  3. $s_j \equiv$ cost shares.

Marshall's Laws

$$
\begin{aligned}
x_i &= y\mu_i(w) \\
&\rightarrow \log x_i = \log y + \log \mu_i(w). \\
&\rightarrow \frac{\partial \log x_i}{\partial w_j} = \frac{\partial \log y}{\partial \log p}\frac{\partial \log p}{\partial w_j} + \frac{\mu_{ij}(w)}{\mu_i(w)} \\
&= -\eta \frac{\partial \log \mu(w)}{\partial w_j} + \frac{\mu_{ij}}{\mu_i} \\
&= -\eta \frac{\mu_j}{\mu} + \frac{\mu_{ij}}{\mu_i} \\
&= -\eta \frac{\mu_j}{\mu} + \frac{\mu_{ij}\mu}{\mu_i\mu_j}\frac{\mu_j}{\mu} \\
&= -\eta \frac{\mu_j}{\mu} + \sigma_{ij}\frac{\mu_j}{\mu} \\
&= (\sigma_{ij} - \eta)\frac{\mu_j}{\mu} = (\sigma_{ij} - \eta)\frac{x_j}{\mu} \rightarrow w_j \frac{\partial \log x_i}{\partial w_j} = (\sigma_{ij} - \eta)\underbrace{\frac{w_j x_j}{\mu}}_{s_j} = \epsilon_{ij} - \eta s_j
\end{aligned}
$$

Marshall's Laws

- Key relationship:

$$\frac{\partial \log x_i}{\partial \log w_j} = \underbrace{\epsilon_{ij}}_{\text{Substitution Effect}} - \underbrace{\eta s_j}_{\text{Scale Effect}} \tag{22}$$

- Substitution Effect: how much does demand for input x_i increases for an increase in the price of input x_j, *holding y fixed*.

- Scale Effect: to understand this, note that since $\mu(w) = p$

$$\begin{aligned} \frac{\partial \log p}{\partial w_j} &= \frac{\mu_j(w)}{\mu(w)} \\ \frac{w_j \partial \log p}{\partial w_j} &= \frac{w_j \mu_j(w) y}{\mu(w) y} = \frac{w_j x_j}{C} = s_j \end{aligned} \tag{23}$$

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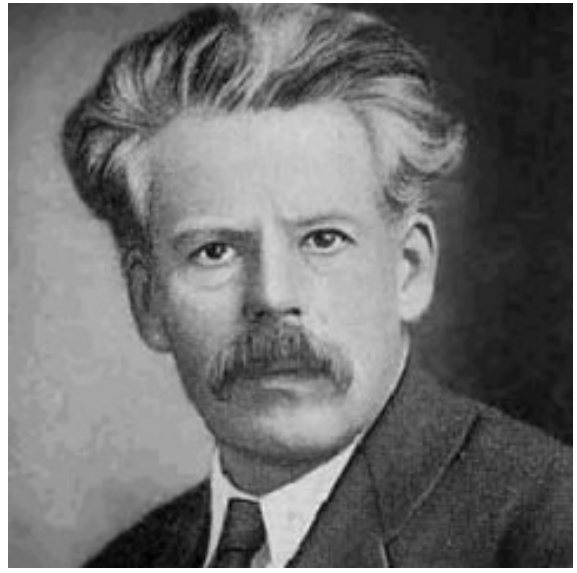
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- Does this bring back any memories?



Discussion

- Neo-classical labor demand framework $\rightarrow$ national or industry wide changes to the wage structure.
- Firms: all have CRS technology and are price-takers in the input market.
- Number and size of the firms is indeterminant.

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- Neo-classical labor demand framework $\rightarrow$ national or industry wide changes to the wage structure.
- Firms: all have CRS technology and are price-takers in the input market.
- Number and size of the firms is indeterminant.
- Very different from other fields!
- IO, Trade: lots of firm level heterogeneity in productivity. Does this productivity heterogeneity "spills" into wages?
- Topic of the last two lectures.

Getting ready for war

- Useful benchmark is the Cobb-Douglas CRS benchmark ($\sigma = 1$)

$$y = f(K, L) = AL^{\alpha}K^{(1-\alpha)} \tag{24}$$

- Note that:

$$\begin{aligned} w &= \alpha A L^{\alpha-1} K^{1-\alpha} \\ wL &= \alpha A L^{\alpha} K^{1-\alpha} \\ \frac{wL}{y} &= \alpha \end{aligned} \tag{25}$$

- α is the labor share (how much of GDP goes to labor?).
  - An entire generation of economists thought that $\alpha = 0.66$.
  - Actually declining over the last years (active area of research).
- Starting point to understand productivity trends:

$$\underbrace{\frac{y}{L}}_{\text{Average Product}} = A(\frac{K}{L})^{1-\alpha} \tag{26}$$

- Therefore in the C-D marginal product is just average product times a constant

$$w = \frac{\partial y}{\partial L} = \alpha \frac{y}{L} \tag{27}$$

Labor Share is declining in many places...

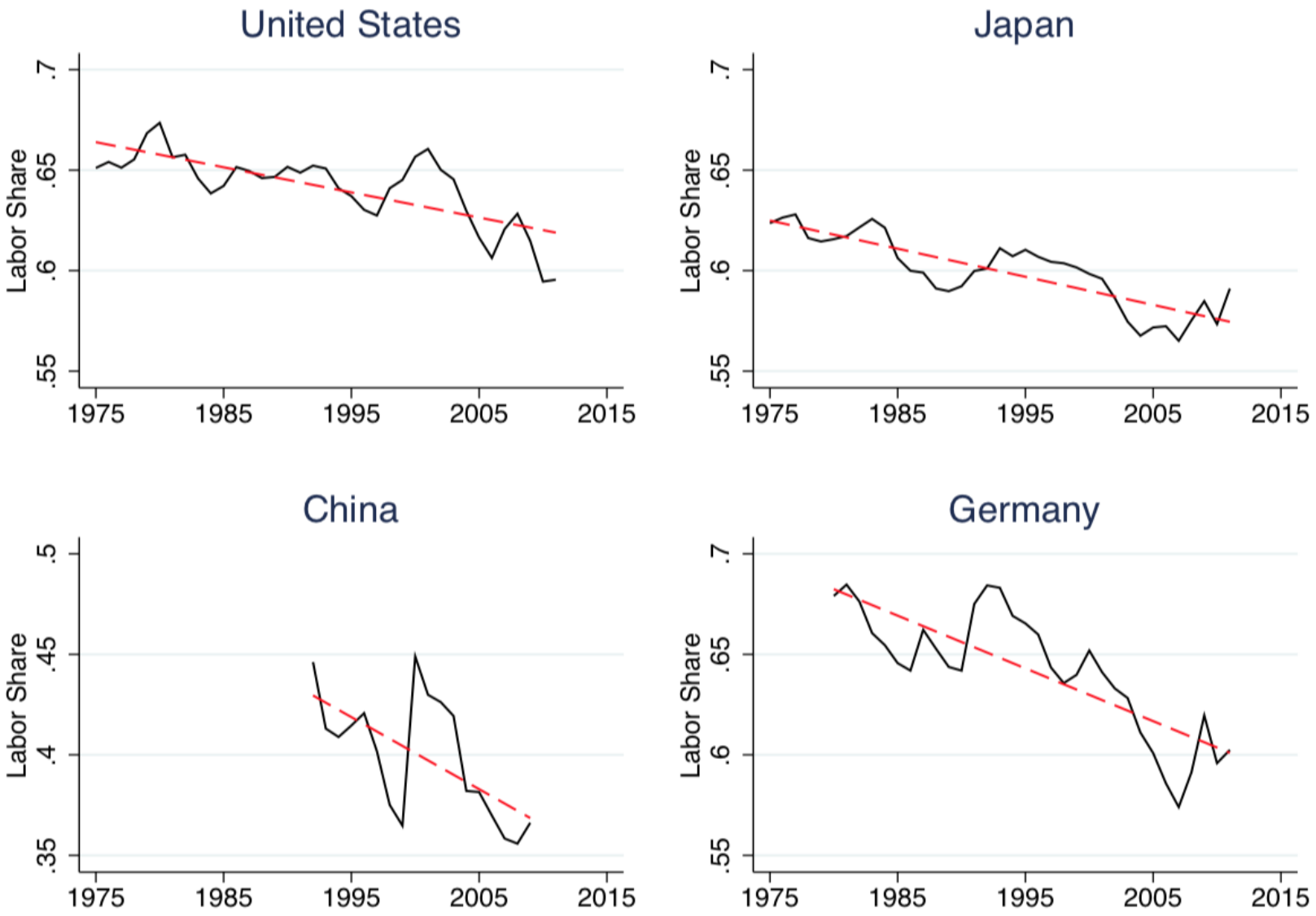


Figure 2: Declining Labor Share for the Largest Countries

Something deep is happening now....

